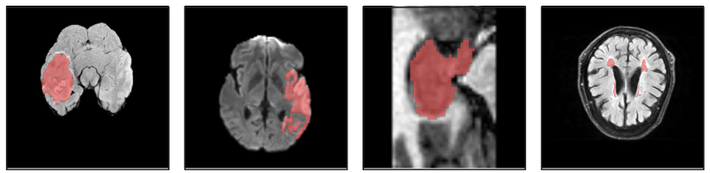
Which adaptation strategy for a new task? A rule that is knowable before training: does the task inherit a large head?

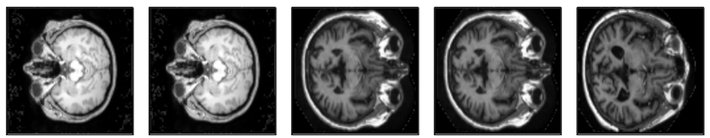
(a) Nine tasks, six cohorts, 32 labelled volumes each

Large task head: inherits the 8.45M-parameter decoder



BraTS glioma | ISLES stroke | Hippocampus | WMH, 5 scanners

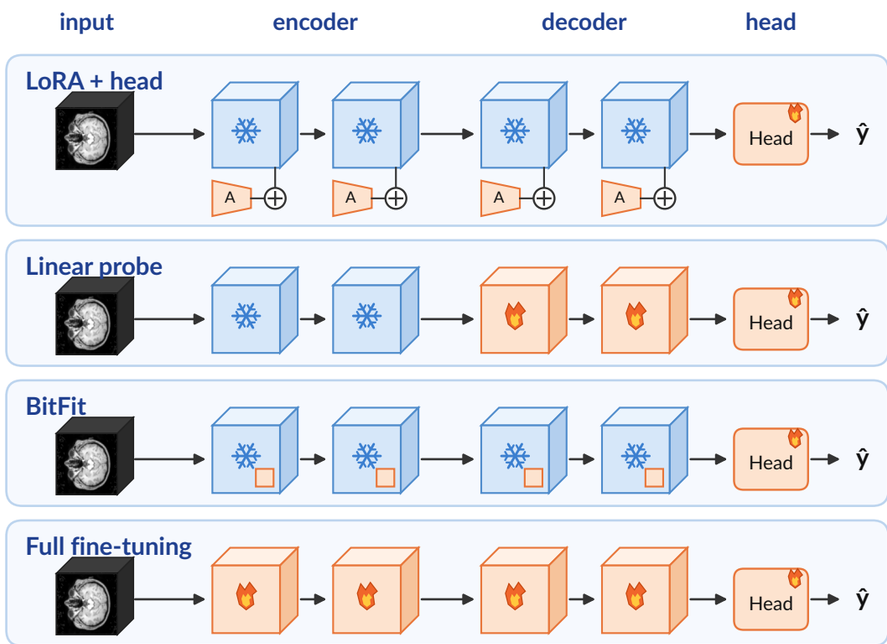
Small task head: ~1K-parameter head on pooled features



IXI age | IXI sex | OASIS age | OASIS sex | OASIS CDR

One held-out test slice per task; segmentation targets in red. Six cohorts, five scanners; the head size is a property of the task that is known before any training run.

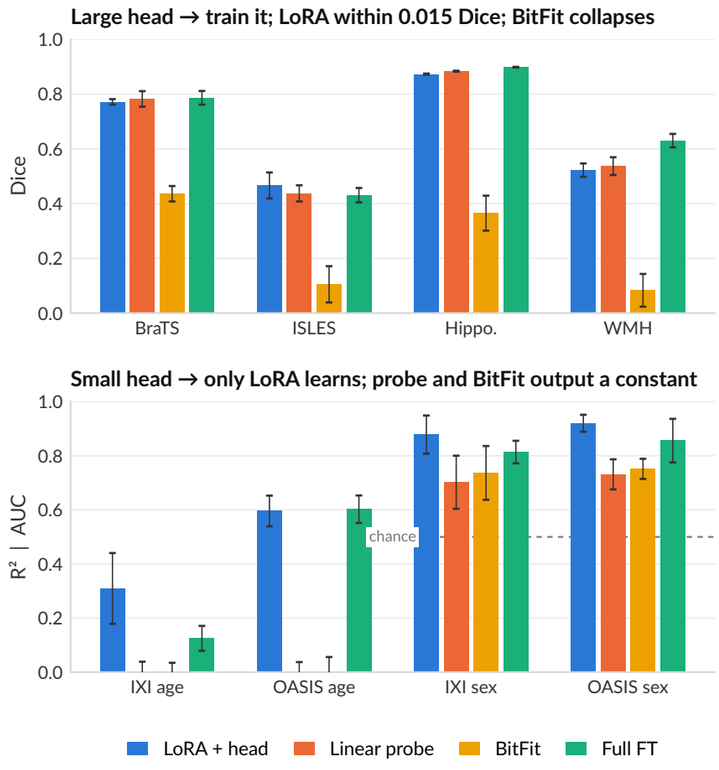
(b) Adaptation strategies: what is frozen and what is trained



Input: 3D MRI, C × 64³ patch (96³ whole head for age / sex). Backbone: FOMO 3D U-Net; 32 volumes, 100 epochs, one recipe.

= frozen | = trained for the new task | A = low-rank adapter (LoRA) | = trainable bias (BitFit)

(c)



Forgetting is symmetric: a second segmentation task erases the first in either order (BraTS 0.78→0.20, ISLES 0.40→0.01); frozen strategies cannot forget.

Pretraining is task-dependent: ΔDice +0.000 BraTS, −0.030 ISLES, +0.024 WMH vs random init; run a random-init control.

Protocol: subject-grouped splits, whole-volume inference, 5 seeds, 346 runs, paired per-case tests with Holm correction.